~~Extapolare~~

2017 - CD - Final - A

Notatii PGP: $M - N_1(h) = \underbrace{k_1 h + k_2 h^2 + k_3 h^3 + \dots}_{O(h)}$

$\Rightarrow M - N_1(h) \simeq k_1 h$

We need higher order

1. $M = N_1(h) + k_1 h^2 + k_2 h^4 + k_3 h^6 + \dots$ (1)

$M = N_1(h/2) + k_1 (h/2)^2 + k_2 (h/2)^4 + k_3 (h/2)^6 + \dots$ (2)

$2 * (2) - (1) \Rightarrow$

$3M = 4N_1(h/2) - N_1(h) + k_2 \frac{h^4}{4} - k_2 h^4 + \dots$

~~$M = \frac{4}{3} N_1(h/2) - \frac{1}{3} N_1(h) + \dots$~~

$M = 4/3 N_1(h/2) - \frac{N_1(h)}{3} + \tilde{k}_2 h^4 + \tilde{k}_3 h^6 + \dots$

$\Rightarrow N_2(h) = 4/3 N_1(h/2) - N_1(h)/3$

2. ~~function [a0, a1~~

Nu vreau să mi se facă pe de rost. Ce fac?

$$E = E_2(a_0, a_1, a_2) = \sum_{i=1}^{m} [y_i - (a_2 x_i^2 + a_1 x_i + a_0)]^2$$

$$\frac{\partial E}{\partial a_0} = 0 \Rightarrow 2 \sum_{i=1}^{m} (y_i - a_2 x_i^2 - a_1 x_i - a_0)(-1) =$$

$$\frac{\partial E}{\partial a_1} = 0 \Rightarrow 2 \sum_{i=1}^{m} (y_i - a_2 x_i^2 - a_1 x_i - a_0)(-x_i) = 0$$

$$\frac{\partial E}{\partial a_2} = 0 \Rightarrow 2 \sum_{i=1}^{m} (y_i - a_2 x_i^2 - a_1 x_i - a_0)(-x_i^2) = 0$$

$$\Rightarrow \begin{cases} a_0 \cdot m + a_1 \cdot \sum_{i=1}^{m} x_i + a_2 \cdot \sum_{i=1}^{m} x_i^2 = \sum_{i=1}^{m} y_i \\ a_0 \sum_{i=1}^{m} x_i + a_1 \sum_{i=1}^{m} x_i^2 + a_2 \cdot \sum_{i=1}^{m} x_i^3 = \sum_{i=1}^{m} x_i y_i \\ a_0 \sum_{i=1}^{m} x_i^2 + a_1 \sum_{i=1}^{m} x_i^3 + a_2 \sum_{i=1}^{m} x_i^4 = \sum_{i=1}^{m} x_i^2 y_i \end{cases}$$

$$A = \begin{bmatrix} m & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{bmatrix}$$

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function [a0, a1, a2] = cmmp(x, y)
~~Am~~ = length(x);
A = [ m  sum(x)   sum(x.^2);
      sum(x)  sum(x.^2)  sum(x.^3);
      sum(x.^2)  sum(x.^3)  sum(x.^4) ];
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= [sum(y) ;  sum(x.* y) ;  sum(x.^2 .* y)];

[a0 , a1 , a2] = (A \ b);
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Ex. 3) $\int_0^1 f(x)\,dx \simeq C_0 f(0) + (C_0 + C_1) f(x_1)$

$N=1$, 2 radăcini $\Rightarrow m = 2\cdot 2 - 1 = 3 \equiv$ grad de validare.

Verificăm $f(x) = 1, x, x^2, x^3$

$f(x) = 1$: $\int_0^1 f(x)\,dx = \int_0^1 1\,dx = 1$

$C_0 f(0) + (C_0 + C_1) f(x_1) = 2C_0 + C_1$

$\Rightarrow 2C_0 + C_1 = 1$ (1)

$f(x) = x$: $\int_0^1 f(x)\,dx = \int_0^1 x\,dx = \frac{x^2}{2}\Big|_0^1 = \frac{1}{2}$ (2)

$\Rightarrow (C_0 + C_1) x_1 = \frac{1}{2}$

$f(x) = x^2$: $\int_0^1 f(x)\,dx = \int_0^1 x^2\,dx = \frac{x^3}{3}\Big|_0^1 = \frac{1}{3}$ (3)

$\Rightarrow (C_0 + C_1) x_1^2 = \frac{1}{3}$

$f(x) = x^3$: $\int_0^1 f(x)\,dx = \int_0^1 x^3\,dx = \frac{x^4}{4}\Big|_0^1 = \frac{1}{4}$ (4)

$(C_0 + C_1) x_1^3 = \frac{1}{4}$

(3)/(2) $\Rightarrow x_1 = \frac{2}{3}$

(4)/(3) $\Rightarrow x_1 = \frac{4}{3} \Rightarrow$ Nu operăm la $f(x) = x^2$

$\Rightarrow \boxed{m = 2}$

$\left.\begin{aligned} C_0 + C_1 &= \frac{1}{2}\cdot\frac{3}{2} = \frac{3}{4} \\ 2C_0 + C_1 &= 1 \end{aligned}\right\} \Rightarrow \begin{cases} C_0 = \frac{1}{4} \\ C_1 = \frac{1}{2} \end{cases}$

$$\int_0^1 f(x)\,dx = \frac{1}{4} f(0) + \left(\frac{1}{4}+\frac{1}{2}\right) f\left(\frac{2}{3}\right)$$

pentru $f(x) \in P_m$, unde $\boxed{m=2}$

Testare $f(x) = 2x^2+3x-2$

$$\int_0^1 2x^2+3x-2\,dx = 2\frac{x^3}{3} + 3\frac{x^2}{2} - 2x\Big|_0^1 = \frac{2}{3} + \frac{3}{2} - 2 = \boxed{\frac{1}{6}}$$

$$\frac{1}{4} f(0) + \frac{3}{4} f\left(\frac{2}{3}\right) = \frac{1}{4}(-2) + \frac{3}{4}\left(2\cdot\left(\frac{2}{3}\right)^2 + 3\cdot\frac{2}{3} - 2\right) =$$

$$= -\frac{1}{2} + \frac{3}{4}\left(2\cdot\frac{4}{9} + 2 - 2\right) = \boxed{\frac{1}{6}}$$

Aproximări Padé — polinoame care calculează mai ușor decât fct. complexe

- se folosesc pentru a aproxima o funcție pe un interval închis I care îl conține pe 0

$$r(x) = \frac{p(x)}{q(x)} = \frac{p_0 + p_1 x^1 + p_2 x^2 + \dots + p_n x^n}{q_0 + q_1 x^1 + q_2 x^2 + \dots + q_m x^m}$$

$r(x)$ ↙ approx. Padé

$\frac{p(x)}{q(x)}$ ↳ approx. rațională

$N = n + m \equiv$ gradul aproximării

Pentru a găsi $r(x)$ pentru $f(x)$ trebuie ca toți coeficienții lui x^k, cu $k = 0, 1, \dots, N$ din ecuația:

$$(\text{seria Maclaurin a lui } f(x))(1 + q_1 x + q_2 x^2 + \dots + q_m x^m) - (p_0 + \dots + p_n x^n)$$

↳ îl fixăm 1 ca să fim siguri că r este definit în 0

să fie 0.

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4. $N = 2 = n + m$

(I) $n = 0, \quad m = 2$

$$\left(1 - x + \frac{x^2}{2} + \dots\right)\left(1 + q_1 x + q_2 x^2\right) - (p_0) =$$

$$= 1 + q_1 x + q_2 x^2 - x - q_1 x^2 + \frac{x^2}{2} + \dots - p_0 =$$

$$= \underbrace{1 - p_0}_{x^0} + \underbrace{q_1 x - x}_{x^1} + \underbrace{q_2 x^2 - q_1 x^2 + \frac{x^2}{2}}_{x^2}$$

$$\begin{cases} 1 - p_0 = 0 & \Rightarrow p_0 = 1 \\ q_1 - 1 = 0 & \Rightarrow q_1 = 1 \\ q_2 - q_1 + \frac{1}{2} = 0 & \Rightarrow q_2 = 1 - \frac{1}{2} = \frac{1}{2} \end{cases}$$

$$\Rightarrow R(x) = \frac{1}{1 + x + \frac{x^2}{2}}$$

$f(x) = e^{-x}$, $f(0.5) = 0.60653$

$R(0.5) = 0.61538$

(II) $n = 1, \quad m = 1$

$$\left(1 - x + \frac{x^2}{2} + \dots\right)\left(1 + q_1 x\right) - \left(p_0 + p_1 x\right) =$$

$$= 1 + q_1 x - x - q_1 x^2 + \frac{x^2}{2} - p_0 - p_1 x$$

$$= \underbrace{1 - p_0}_{x^0} + \underbrace{q_1 x - x - p_1 x}_{x^1} - \underbrace{q_1 x^2 + \frac{x^2}{2}}_{x^2}$$

$$\left.\begin{cases} 1 - p_0 = 0 \Rightarrow p_0 = 1 \\ q_1 - 1 - p_1 = 0 \Rightarrow p_1 = \frac{1}{2} - 1 = -\frac{1}{2} \\ -q_1 + \frac{1}{2} = 0 \Rightarrow q_1 = \frac{1}{2} \end{cases}\right\} \Rightarrow R(x) = \frac{1 - \frac{1}{2}x}{1 + \frac{1}{2}x}$$

$R(0.5) = 0.6$

~~$n=0, w=2$~~

~~$\left(1-x+\frac{x^2}{2}+\dots\right)\left(1+q_1x+q_2x^2\right)$~~

$n=2, w=0$

$$\left(1-x+\frac{x^2}{2}+\dots\right)(1)-(p_0+p_1x+p_2x^2)=$$

$$=1-x+\frac{x^2}{2}-p_0-p_1x-p_2x^2=$$

$$=\underbrace{1-p_0}_{x^0}\underbrace{-x-p_1x}_{x^1}+\underbrace{\frac{x^2}{2}-p_2x^2}_{x^2}$$

$$\begin{cases} 1-p_0=0 \Rightarrow p_0=1 \\ -1-p_1=0 \Rightarrow p_1=-1 \\ \frac{1}{2}-p_2=0 \Rightarrow p_2=\frac{1}{2} \end{cases}$$

$$\Rightarrow \pi(x)=1-x+\frac{x^2}{2}$$

$$\pi(0.5)=0.625$$